

Matthew Ju

matthewju.com | matthew.ju@outlook.com | github.com/matthewju06 | linkedin.com/in/matthew-ju

Education

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science & Statistics — GPA: 3.80/4.00

Expected May 2027

Champaign, Illinois

Projects

ListingLab - Market Analytics Platform (listing-lab.vercel.app)

December 2025 – Present

Python, Flask, OAuth 2.0, Chart.js, HTML/CSS/JS

- Designed, built, and deployed a public full-stack web application on Vercel that aggregates live eBay Browse API data and applies IQR-based outlier filtering
- Developed an interactive analytics dashboard with JavaScript and Chart.js to visualize pricing distributions, condition breakdowns, and time-series trends not available in standard eBay search
- Implemented pagination for up to 1,000 items per query, price range filtering, and persistent user search history
- Created to solve market research inefficiencies while running an eBay store with \$200k+ in sales over 6 years

MCP Weather Server

January 2026

Python, FastMCP, Docker, Open-Meteo API

- Built a Python-based MCP (Model Context Protocol) server to deliver live weather data to LLM chats and agents, providing standardized endpoints for conditions, forecasts, and alerts
- Implemented 48,000+ city geolocation lookup and integrated the Open-Meteo REST API; containerized the service with Docker for reproducible deployment

Locked-In Toolbox (Course Project)

October 2025 – December 2025

React, TypeScript, Supabase, SQL, REST APIs

- Developed a full-stack productivity application with React, using a Supabase backend for secure data storage and authentication
- Created UI components to visualize user study activity and calendar insights through interactive metric dashboards

Technical Experience

AbbVie

June 2024 – August 2024

Data Engineering Intern

North Chicago, Illinois

- Built a real-time manufacturing data pipeline reading sensor data from industrial equipment over ethernet and wireless networks into Python-based processing workflows
- Implemented streaming signal processing in Python (Pandas) using moving average and median filters to denoise sensor signals and detect experimental anomalies in near real time
- Automated data extraction and visualization, reducing experimental processing time by 50%

Skills

Programming Languages: Python, C++, JavaScript, Java, SQL, HTML, CSS

Frameworks & Tools: Git, Docker, MCP (FastMCP), REST APIs, Flask, OAuth 2.0, Supabase (PostgreSQL), Vercel, React, Chart.js, Pandas, NumPy, scikit-learn, Matplotlib

Coursework: Data Structures & Algorithms, Software Design, Linear Algebra, Computer Architecture, Numerical Methods, Probability & Statistics II

Additional Experience

YJ Consulting

May 2025 – Present

Co-Founder

Vernon Hills, Illinois

- Led client restaurant website modernization and digital menu optimization, increasing viewership by 37% and sales by 8%; built company website: <https://yjconsulting.pro/>

Champaign Urbana Business and Engineering (CUBE) Consulting

February 2025 – Present

Technical Consultant

Champaign, Illinois

- Conducted feasibility analysis for a student startup's SaaS platform to define technical and financial requirements
- Led website planning for an on-campus cafe